# The Claim Verification Pipeline: A Methodology for Auditing AI-Generated Mathematics

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**Abstract**

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating heuristic solutions to complex scientific problems, including combinatorial mathematics. However, the probabilistic, non-deterministic nature of these models introduces severe risks of hallucination, definitional tautology, and subtle logical errors. We present the *Claim Verification Pipeline*, a robust methodology developed within the SocrateAI Agora framework. This pipeline imposes a deterministic, four-gate standard for evaluating AI-generated mathematical claims: rigorous literature grounding, large-scale computational falsification, zero-axiom formalization in Lean 4, and peer-style critical review. Through an investigation into AI-generated bounds for hypercube tensor rank and exact Ramsey numbers ($R(3,3,4)$ and $R(4,6)$), we demonstrate how a multi-agent system can autonomously separate "wishful thinking" from genuine mathematical contribution. We articulate a new AI-human philosophy that embraces generative creativity while enforcing absolute verification.

## 1 Introduction

The integration of Artificial Intelligence into mathematical research has bifurcated into two distinct paradigms: the symbolic, deterministic verification engines (e.g., Lean 4, Coq) and the probabilistic, generative heuristic engines (e.g., Large Language Models like DeepSeek-Prover, Gemini, and GPT-4). While the former guarantees correctness at the kernel level, it lacks intuition; while the latter mimics intuition and creative leap-making, it suffers from severe hallucination and lacks rigorous constraints.

The challenge for mathematicians and engineers is not merely to build better models, but to construct systems that safely harness the creativity of LLMs while mitigating their philosophical risks. The danger lies in human bias: researchers, eager for breakthroughs, can fall prey to "wishful thinking," accepting AI-generated plausibility as proof.

In this paper, we detail the *Claim Verification Pipeline*, an autonomous methodology implemented by the SocrateAI Agora. We apply this pipeline to claims generated by an evolutionary LLM system (AlphaEvolve) concerning Ramsey theory and tensor rank on discrete hypercubes.

## 2 The Claim Verification Pipeline

The pipeline operates on a falsification-first principle. Instead of attempting to prove an AI claim, the system employs specialized agents to rapidly falsify it. Only claims that survive falsification proceed to formal verification.

### 2.1 Gate 1: Literature Grounding

The first step is a rigorous audit of the claim against established literature. AI systems frequently hallucinate bounds that contradict known theorems. For example, an AI claim that

$R(4,4,4) = 26$ was instantly dismissed by our pipeline upon discovering Bläser's 2003 theorem proving $R(4,4,4) \geq 28$. Similarly, a claim regarding $R(3,3,4) \in [51,62]$ was corrected to the exact known value $R(3,3,4) = 30$ (Codish et al., 2016).

### 2.2 Gate 2: Computational Falsification

Claims surviving literature review are subjected to high-performance computational falsification. We developed a highly optimized Rust-based Simulated Annealing (SA) solver utilizing bitset adjacency and delta evaluation.

- **Performance**: The Rust solver achieved 2.5 to 21 million steps per second, a $420,000\times$ speedup over equivalent Python implementations.
- **Result**: In testing $R(4,6) \geq 36$, our system executed over 2 billion SA evaluations in under 15 minutes. It found that basic SA consistently stalled at 15 violations for $K_{35}$. This computational bound confirmed that Exoo's 2012 construction required specialized heuristics, and prevented the system from reporting a false positive.

### 2.3 Gate 3: Lean 4 Formalization

The gold standard for mathematical truth is zero-axiom, zero-sorry formalization.

#### 2.3.1 The ExactRationalWitness

The most significant verified contribution of this investigation is the `ExactRationalWitness`. The AI proposed polynomial bounds over the hypercube $H(21,2)$ using Krawtchouk polynomials. While the real-analysis proofs introduced continuous axioms, our pipeline mapped the problem onto the exact rational field $\mathbb{Q}$.

```
/-- ExactRationalWitness: W(w) = P(w)^2 + 1 > 0 -/
def W_alien (w : Nat) : Rat := ...
theorem W_alien_base_pos (w : Nat) (hw : w <= 21) : 0 < W_alien w := by
    ...
```

This shifted the verification from complex analysis to computable $\mathbb{Q}$-decidability, allowing Lean 4's `native_decide` to fully verify the claim at the kernel level.

### 2.4 Gate 4: Peer-Style Critical Review

The final gate is an LLM-driven "dialectical" review by the Socrate agent, which critiques the novelty and significance of the verified claim, guarding against "definitional tautologies" where a statement is true but mathematically trivial.

## 3 A New Science AI-Human Philosophy

The integration of generative Artificial Intelligence into the absolute rigor of mathematics, physics, and engineering necessitates a profound epistemological shift. We propose a genuinely new *Science AI-Human Philosophy* to navigate this frontier, born from our experience with the Alien Mathematics project.

### 3.1 The Power and Peril of Non-Deterministic Engines

Large Language Models (LLMs) are fundamentally non-deterministic, probabilistic engines. Their immense power lies precisely in their lack of rigid, classical logical constraint. They excel at creative leaps, interpolating across vast, high-dimensional latent spaces of human knowledge to propose novel configurations, intuition-driven hypotheses, and "Alien" heuristics.

However, this very creativity is inherently coupled with hallucination. An LLM is not a reasoning engine; it is a plausibility engine. The philosophical risk to scientific rigor emerges when we treat a non-deterministic generator as an oracle of truth. The challenge for the modern inventor, mathematician, and engineer is to harness this chaotic, probabilistic creativity without compromising the deterministic bedrock of the scientific method.

### 3.2 The Psychological Trap: Human Bias and the Loss of Critique

The greatest threat in AI-assisted discovery is not the machine's capacity to hallucinate, but the human's psychological vulnerability to it. Humans are deeply biased creatures. We harbor an intrinsic, profound desire to discover breakthroughs—to find the "Alien Mathematics" that changes the paradigm.

When an AI presents a highly sophisticated, coherent, yet fundamentally flawed mathematical proof or physical theory, it triggers a dangerous psychological trap. The researcher, wanting too much to find the breakthrough, is prone to suspend their critical faculties. They engage in "wishful thinking," becoming a retroactive rationalizer for the AI's hallucination, bending their own logic to make the AI's output true. This loss of critical objectivity is the true philosophical crisis of our era.

### 3.3 The Tripartite Epistemology

To resolve this tension, we mandate a new tripartite epistemology for AI-human scientific collaboration:

**1. The Principle of Probabilistic Inspiration:** The AI must be relegated to the role of the "subconscious mind" of the scientific endeavor. It proposes, dreams, and connects, but it never concludes.

**2. The Principle of Absolute Adversarial Skepticism (Falsification First):** Because humans are vulnerable to wishful thinking, the verification loop must be structurally adversarial. The system must aggressively attempt to destroy the AI's claim via high-performance computational falsification (e.g., our 21M steps/s Rust solver) before any human emotional investment occurs.

**3. The Primacy of the Deterministic Kernel:** Truth is neither established by the eloquence of the LLM nor the consensus of hopeful researchers. Truth is strictly that which compiles in a deterministic, zero-axiom verification kernel (e.g., Lean 4). By mapping continuous, ambiguous problems onto exact, decidable domains (such as the $\mathbb{Q}$-arithmetic of the `ExactRationalWitness`), we insulate the scientific record from both AI hallucination and human bias.

## 4 Conclusion

The SocrateAI Agora investigation into Alien Mathematics did not yield a mathematical breakthrough in Ramsey theory. However, it successfully built a 2.5M steps/s Rust infrastructure and, more importantly, a reusable *Claim Verification Pipeline*. This methodology provides a safe, rigorous framework for integrating the chaotic creativity of modern LLMs into the strict discipline of formal mathematics.